James W. T. Keeble

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London, UK | UK Citizen | English (native), German (B1)

Computational physicist specialising in scientific ML: neural-network architectures and numerical algorithms for quantum many-body systems, scaled on multi-GPU clusters and benchmarked against exact methods.

EXPERIENCE

Independent Researcher*, *in collaboration with Bielefeld University *June 2026 – Present*

- Preparing an open-source release of the multi-GPU eigensolver, with tests, CI and reproducible benchmarks.
- Developing new symmetry-aware neural-network wavefunctions for heavier nuclei on multi-GPU clusters.

Postdoctoral Researcher*, *Bielefeld University, Germany *April 2024 – May 2026*

- Built a multi-GPU neural-network variational Monte Carlo library in JAX, benchmarked against exact FCI where tractable and more accurate than coupled cluster (CCSD, CCSD(T)) on 40-qubit systems.
- Identified quantum-state magic (non-stabilizerness) as a predictor of neural-network accuracy, showing where neural quantum states fail (first-author preprint).
- Engineered an MPI-enabled, multi-GPU Lanczos eigensolver in Julia (CUDA.jl) for nuclear structure, 50× faster than the Fortran solver BIGSTICK, built with an agentic AI-assisted workflow.
- Replaced computationally expensive CPU routines with custom CUDA.jl kernels, enabling the research group to scale calculations from 24-qubit to 44-qubit systems.
- Maintained collaborative research software with CI/CD, automated testing and Apptainer containers.

Doctoral Researcher*, *University of Surrey, UK *October 2018 – April 2023*

- Developed an antisymmetric neural wavefunction in PyTorch for fermions in continuous space, outperforming Hartree–Fock and configuration interaction at fixed basis size across interaction strengths.
- Created a neural-network wavefunction for the deuteron in continuous momentum space, among the first neural-network quantum states for an atomic nucleus.
- Mentored PhD students in scientific ML software development, coding best practice and HPC workflows.

OPEN SOURCE & COMMUNITY

- Led the implementation of Decisional Gradient Descent, a game-theory extension of stochastic reconfiguration that outperformed it, published in *Phil. Trans. R. Soc. A*.
- Evaluated functorch per-sample-gradient transforms, giving PyTorch developers validation results and usage examples ahead of their integration into torch.func.
- PyTorch Foundation Pace-Setter Award (2023) for nearly a hundred accepted forum solutions.

EDUCATION

- PhD, Theoretical Nuclear Physics, University of Surrey, UK | Graduated April 2023
- MPhys, Physics (First-Class Honours), University of Surrey, UK | Graduated October 2018

PUBLICATIONS

- **J. W. T. Keeble**, A. Lovato, C. E. P. Robin, “Neural Quantum States in Non-Stabilizer Regimes: Benchmarks with Atomic Nuclei,” *arXiv:2603.28646* (2026), preprint.
- M. Drissi, **J. W. T. Keeble**, J. Rozalén Sarmiento, A. Rios, “Second-order Optimisation Strategies for Neural Network Quantum States,” *Phil. Trans. R. Soc. A* 382, 20240057 (2024).
- **J. W. T. Keeble**, A. Rios, “Machine Learning the Deuteron,” *Phys. Lett. B* 809, 135743 (2020).
- Google Scholar (195 citations, h-index 6): scholar.google.com/citations?user=XednZsIAAAAJ

SKILLS

- Scientific ML: neural quantum states, symmetry-aware architectures (Pfaffians, Transformers), variational Monte Carlo, MCMC, natural-gradient optimisation, exact diagonalisation, numerical benchmarking
- Programming: Python, Julia, Fortran, Bash
- GPU and distributed computing: JAX, PyTorch, CUDA.jl, MPI, multi-GPU and multi-node, mixed precision
- Software engineering: Git, CI/CD, automated testing, Apptainer, SLURM, agentic AI development (Claude)